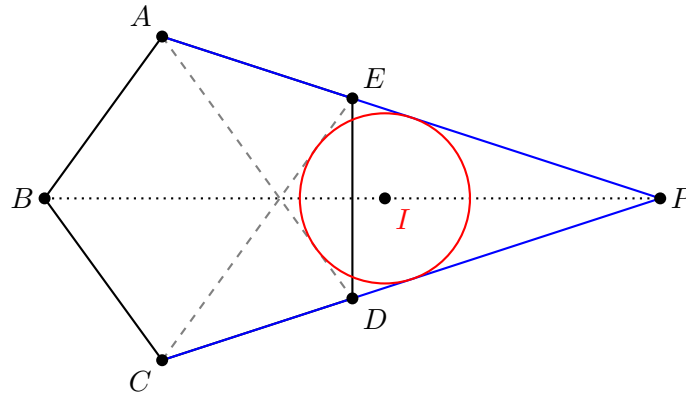


Geometric Functional Equations

Some Techniques Via Solved Examples

Problem 1. Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ be a function on the plane such that for any triangle ABC with incenter I , $f(I) = \frac{f(A)+f(B)+f(C)}{3}$. Prove that f is a constant function.

Solution. We aim to prove that $f(A) = f(B)$ for any two arbitrary points A and B in the plane. Construct a regular pentagon $ABCDE$ using A and B as adjacent vertices. Let the extensions of sides AE and CD intersect at a point P .



A regular pentagon possesses reflectional symmetry. The line passing through B and bisecting ED acts as an axis of symmetry for the pentagon $ABCDE$. Because lines AE and CD are symmetric with respect to this axis, their intersection P must also lie precisely on this axis.

Thus, the line BP is the angle bisector of $\angle P$ in $\triangle ADP$. Since the incenter of any triangle lies on its angle bisectors, the incenter I of $\triangle ADP$ must lie on the line BP .

If we reflect the plane across the axis BP :

- Point P remains fixed.
- Point A reflects to point C .
- Point D reflects to point E .

This reflection perfectly maps $\triangle ADP$ onto $\triangle CEP$. Because the incenter I lies on the axis of reflection, it remains invariant. Therefore, $\triangle ADP$ and $\triangle CEP$ share the exact same incenter I .

Applying the problem's condition, $3f(I) = f(v_1) + f(v_2) + f(v_3)$, to both triangles:

$$\text{For } \triangle ADP : \quad 3f(I) = f(A) + f(D) + f(P)$$

$$\text{For } \triangle CEP : \quad 3f(I) = f(C) + f(E) + f(P)$$

Equating the two expressions and canceling $f(P)$ yields:

$$f(A) + f(D) = f(C) + f(E) \quad \text{— (Equation 1)}$$

This establishes a functional relation for the vertices of any regular pentagon. Due to the rotational symmetry of the regular pentagon, we can cyclically shift our sequence of vertices. By shifting the starting vertex from A to E (viewing the sequence as E, A, B, C, D), the exact same property must hold:

$$f(E) + f(C) = f(B) + f(D) \quad \text{--- (Equation 2)}$$

Notice that the right side of Equation 1 perfectly matches the left side of Equation 2. Linking them gives:

$$f(A) + f(D) = f(B) + f(D)$$

Subtracting $f(D)$ from both sides, we conclude:

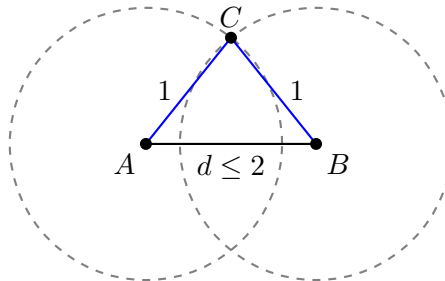
$$f(A) = f(B)$$

Since A and B were arbitrary points used to construct the initial pentagon, the function evaluates to the same value for any two points in the plane. Thus, f is a constant function.

Problem 2. Find all functions $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ such that $f(A) = f(B)$ for any two points A, B in the plane separated by a distance of exactly 1.

Solution. We will prove that f must be a constant function. Let A and B be any two points in the plane such that the distance between them is $d \leq 2$.

Because $d \leq 2$, circles of radius 1 centered at A and B must intersect at some point C (or be tangent if $d = 2$). By definition, the distance $AC = 1$ and the distance $BC = 1$.

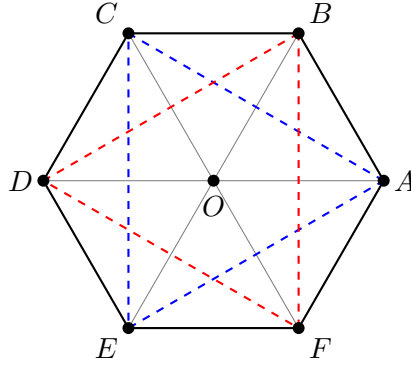


From the given condition, since $AC = 1$, we have $f(A) = f(C)$. Similarly, since $BC = 1$, we have $f(B) = f(C)$. Therefore, $f(A) = f(B)$ for any two points separated by a distance $d \leq 2$.

For any two arbitrary points $X, Y \in \mathbb{R}^2$ separated by a distance $D > 2$, we can construct a sequence of points $X = P_0, P_1, P_2, \dots, P_n = Y$ strictly along the line segment XY such that the distance between any consecutive pair P_i, P_{i+1} is less than or equal to 2. By our previous deduction, $f(P_0) = f(P_1) = \dots = f(P_n)$. Thus, $f(X) = f(Y)$ for all X, Y in the plane, proving f is a constant function.

Problem 3. Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ be a function such that for every equilateral triangle ABC in the plane, $f(A) + f(B) + f(C) = 0$. Prove that $f(P) = 0$ for all points P in the plane.

Solution. Let O be an arbitrary point in the plane. Construct a regular hexagon $ABCDEF$ centered at O .



The hexagon can be divided into six smaller equilateral triangles sharing the vertex O : $\triangle OAB$, $\triangle OBC$, $\triangle OCD$, $\triangle ODE$, $\triangle OEF$, $\triangle OFA$. Applying the given condition to all six triangles and summing the results, we obtain:

$$\begin{aligned} & (f(O) + f(A) + f(B)) + (f(O) + f(B) + f(C)) + \cdots + (f(O) + f(F) + f(A)) = 0 \\ \Rightarrow & 6f(O) + 2(f(A) + f(B) + f(C) + f(D) + f(E) + f(F)) = 0 \quad \text{--- (Equation 1)} \end{aligned}$$

Observe that the vertices of the hexagon also form two larger inscribed equilateral triangles: $\triangle ACE$ (blue) and $\triangle BDF$ (red). By the problem's condition:

$$f(A) + f(C) + f(E) = 0 \quad \text{and} \quad f(B) + f(D) + f(F) = 0$$

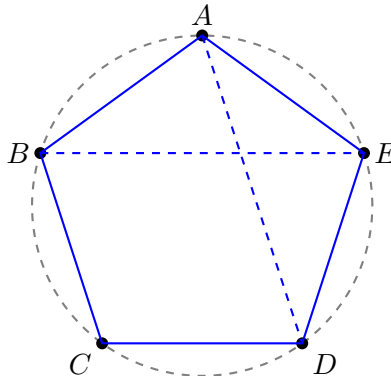
Adding these two equations yields $f(A) + f(B) + f(C) + f(D) + f(E) + f(F) = 0$. Substituting this back into Equation 1, we get:

$$6f(O) + 2(0) = 0 \implies f(O) = 0$$

Since the center O was chosen entirely arbitrarily, $f(P) = 0$ for every point P in the plane.

Problem 4. Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ be a function such that for any four concyclic points A, B, C, D , we have $f(A) + f(B) + f(C) + f(D) = 0$. Prove that $f(P) = 0$ for every point P .

Solution. Consider any circle ω in the plane. Inscribe a regular pentagon $ABCDE$ within this circle.



Because all five vertices lie on the same circle ω , any combination of four vertices forms a concyclic quadrilateral. Applying the given condition to the quadrilaterals $ABCD$ and $BCDE$:

$$\begin{aligned} f(A) + f(B) + f(C) + f(D) &= 0 \\ f(B) + f(C) + f(D) + f(E) &= 0 \end{aligned}$$

Subtracting the second equation from the first yields $f(A) - f(E) = 0$, or $f(A) = f(E)$. Due to the rotational symmetry of the regular pentagon, we can deduce that $f(A) = f(B) = f(C) = f(D) = f(E)$. Let this common value be k .

Because $ABCD$ is a concyclic quadrilateral, $f(A) + f(B) + f(C) + f(D) = 0$, which means $4k = 0 \implies k = 0$. Thus, f evaluates to 0 for any point on the circle ω .

Since ω was an arbitrary circle, and any point P in the plane lies on infinitely many circles, $f(P) = 0$ for all $P \in \mathbb{R}^2$.

Problem 5. Find all continuous functions $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ such that for any two points A, B with midpoint M , the function satisfies:

$$f(A) + f(B) = 2f(M)$$

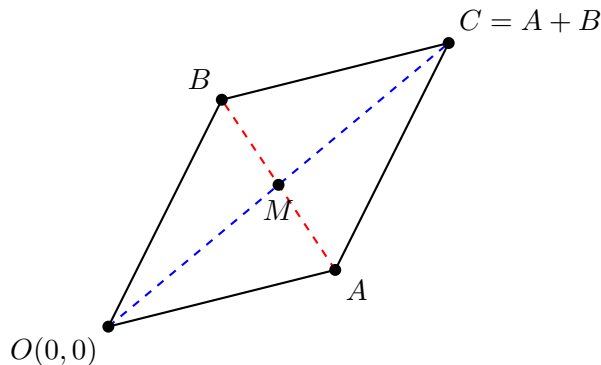
Solution. Let $c = f(0, 0)$ be the value of the function at the origin. We define a new function $g(P) = f(P) - c$. This linear shift ensures $g(0, 0) = 0$. Since $f(A) + f(B) = 2f(M)$, subtracting $2c$ from both sides gives:

$$g(A) + g(B) = 2g(M)$$

Let B be the origin $(0, 0)$. The midpoint of A and the origin is $A/2$. The equation becomes:

$$g(A) + g(0, 0) = 2g(A/2) \implies g(A) = 2g(A/2)$$

Now, let A and B be arbitrary points, and let O be the origin. Construct a parallelogram $OACB$ such that $C = A + B$. The diagonals of a parallelogram bisect each other, meaning the midpoint of OC is identical to the midpoint of AB . Let this intersection point be M .



Applying the midpoint rule to the diagonal AB , we have $2g(M) = g(A) + g(B)$. Applying the midpoint rule to the diagonal OC , we have $2g(M) = g(O) + g(C) = 0 + g(A + B)$. Equating these expressions yields:

$$g(A) + g(B) = g(A + B)$$

This is Cauchy's functional equation in two dimensions. Because f is given to be continuous, g must also be continuous. The only continuous solutions to Cauchy's equation are linear maps. Therefore, $g(x, y) = \alpha x + \beta y$ for some real constants α, β .

Substituting back $f(x, y) = g(x, y) + c$, we conclude that the family of solutions is all linear planes in $\mathbb{R}^3$:

$$f(x, y) = \alpha x + \beta y + c$$

where $\alpha, \beta, c \in \mathbb{R}$.
